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Question

To better understand the burrowing habits of *Alpheus bellulus* (the tiger pistol shrimp), some studies have used resin casting to obtain precise measurements of the shrimps' burrows. Resin casting involves completely filling an empty burrow with a liquid plastic that hardens to create a three-dimensional model; however, recovering the model inevitably requires destroying the burrow. In their 2022 study, Miyu Umehara and colleagues discovered that an x-ray computed tomography (CT) scanner can accurately record a burrow's measurements both at a moment in time and throughout the entire burrow-building process, something that's impossible with resin casting because _____

Which choice most logically completes the text?

Answer

- A. it can only be used on burrows below a certain size.
- B. it does not allow for multiple castings of the same burrow over time.
- C. the casting process takes more time than *A. bellulus* takes to construct a burrow.
- D. the process of recovering the model distorts the resin's shape.